Customer Segmentation

Credit Card Users Unsupervised ML

Segmenting credit card customers based on their usage behavior to uncover actionable business insights.

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Data Preprocessing

Handled Missing Data

Imputed 313 MINIMUM_PAYMENTS (set to 0 for users with 0 payments, used IterativeImputer regression for the remaining 73). Filled 1 missing limit.

Addressed Inconsistencies

Capped the CASH_ADVANCE_FREQUENCY feature to a strict maximum of 1.0, correcting entries that originally exceeded this threshold.

Outlier Handling

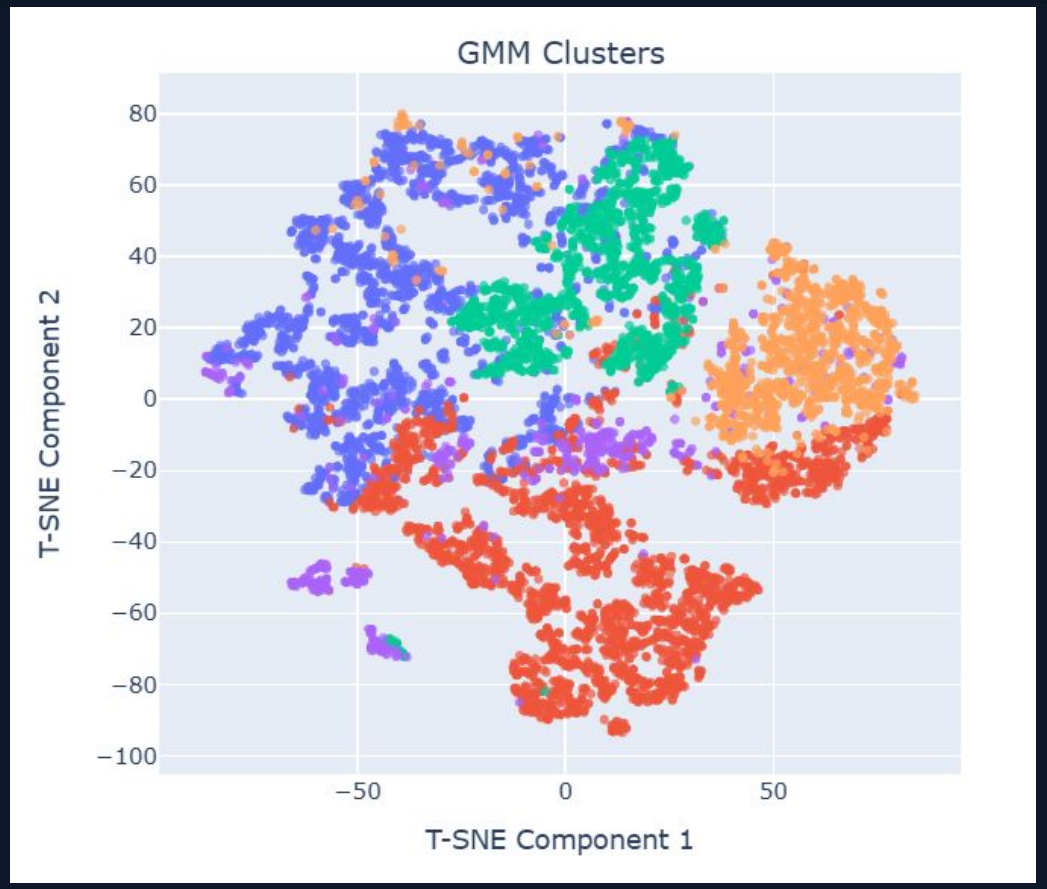
Retained data outliers intentionally, as they accurately represent the critical high-value, heavy-spending customer segment rather than errors.

Data Transformations

Applied log1p to right-skewed features managing large variations, and utilized RobustScaler for model standardization resilient to extreme spenders.

Model Selection & Reduction

- **Dimensionality Reduction:** Evaluated T-SNE into 2D and 3D spaces, visually indicating the presence of 4 to 5 structural clusters.
- **K-Means Validation:** Applied the Elbow Method (WCSS), mapping curve flattening accurately around k=4 and k=5 limits.
- **Final Selection:** Selected Gaussian Mixture Models (GMM) with 5 clusters over traditional K-Means prediction (k=5).
- **Conclusion:** GMM delivered more logical member distributions and handles variance dynamically.



Cluster 0: Low Payers

RISK WATCH CATEGORY

- **Data Profile:** Low payments and minimum payments, but mid-to-high cash advance usage. Maintained mid-level purchases.
- **Behavior Insights:** Customers in this cluster pay very little compared to others and rely on cash advances.
- **Conclusion:** These attributes strongly indicate they might be financially stressed, requiring careful monitoring.

TOP VALUE CUSTOMERS

Cluster 1: Heavy Users

Data Profile: High purchases, transactions, balance, payments, and cash advance usage.

Behavior Insights: Very active processing high spend. Heavy cash advance use presents a dual profile of high-profit but potential high-risk.

Cluster 2: Transactors

HEALTHY ACCOUNTS

- **Data Profile:** Very low balance carried over, low cash advance and minimum payments, combined with mid-level purchases.
- **Behavior Insights:** Customers who maintain consistently low balances, avoid cash advances completely, and pay on time.
- **Conclusion:** Represents highly healthy, low-risk customer behavior with stable ongoing interactions.

TARGETED SPENDERS

Cluster 3:
One-Off Buyers

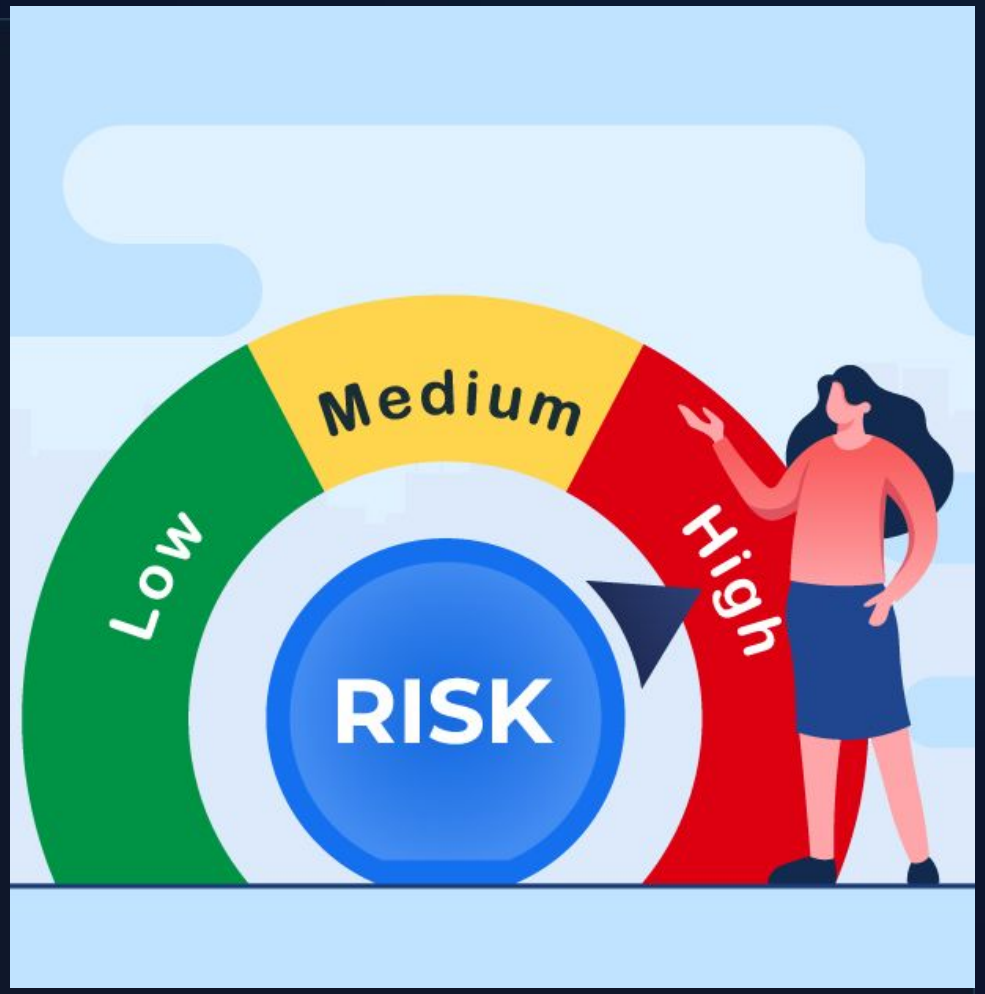
Data Profile: High one-off purchases, mid balance, low cash advance.

Behavior Insights: Customers who prefer making big-ticket, single purchases (like travel or electronics) rather than regular small everyday transactions.

Cluster 4: Cash Driven

HIGHER RISK SEGMENT

- **Data Profile:** Extremely high cash advances and cash advance transactions. Very low regular purchases and installments.
- **Behavior Insights:** Customers using the credit card almost exclusively as a short-term cash loan tool rather than for shopping.
- **Conclusion:** Indicates a high risk demographic that utilizes banking liquidity over merchant transactions.



Business Recommendations

Segment	Customer Persona	Actionable Business Strategy
Cluster 0	Low Payers	Implement early risk monitoring systems and ensure automated payment reminders are consistently sent out.
Cluster 1	Heavy Users	Target with VIP retention & premium rewards. Consider limit increases and offer structured loan alternatives.
Cluster 2	Transactors	Deploy focused reward campaigns and discounts to incentivize higher transaction volumes and dependency.
Cluster 3	One-Off Buyers	Present attractive installment plans at checkout. Target heavily during seasonal campaigns (holidays, travel).
Cluster 4	Cash Driven	Enforce strict risk controls. Proactively market lower-interest structured installment loans for safe conversion.

Suggested Enhancements

Density Models

Experimenting with Agglomerative Hierarchical Clustering to capture arbitrarily shaped clusters that distance-based models miss.

Alternative Reduction

Assessing PCA (Principal Component Analysis) instead of T-SNE before clustering to test if capturing linear variance yields faster separation.

Automatic Validation

Incorporating the Silhouette Score alongside the K-Means Elbow method for a purely mathematical, automated validation of the 'k' limit.